

Fatima Jinnah Women University



A Case Study / Report

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AI vs Human Decision-Making Analysis: A Case Study

Table of Contents:

1. Executive Summary / Abstract
2. Introduction
3. Problem Statement / Case Background
4. Literature Review / Theoretical Framework
5. Methodology / Approach
6. Analysis and Discussion
7. Ethical Analysis
8. Legal and Regulatory Issues
9. Recommendations / Proposed Solution
10. Conclusion
11. Personal Reflection
12. References
13. Appendices

1. Executive Summary / Abstract:

Artificial Intelligence (AI) systems are increasingly being used in important decision-making areas such as healthcare, finance, education, transportation, and business operations. However, there are still concerns regarding the reliability, fairness, transparency, and ethical implications of AI-generated decisions when compared with human judgment. This case study analyzes the differences between AI and human decision-making patterns using data analytics and machine learning techniques. The project collected and analyzed responses from both human participants and AI systems using real-life decision-making scenarios. The collected data was cleaned, transformed, statistically analyzed, and evaluated using machine learning models such as Logistic Regression, Random Forest, and XGBoost. Additional explainable AI techniques such as SHAP analysis and feature importance analysis were also used to interpret the behavior of the models.

The findings showed that although AI systems can provide fast and consistent decisions, human decision-making still performs better in areas requiring emotional understanding, ethical reasoning, and contextual judgment. The study also identified important ethical concerns such as

algorithmic bias, transparency, accountability, privacy, and fairness. Legal and regulatory aspects including PECA, GDPR, and data protection principles were also examined. This case study highlights the importance of combining human intelligence with AI systems rather than fully replacing human decision-making processes. The report concludes with recommendations for responsible AI implementation, ethical governance, and improved transparency in intelligent systems.

2. Introduction:

Decision-making is one of the most important aspects of human behavior. Humans make decisions daily based on emotions, experience, ethics, intuition, and contextual understanding. With the rapid development of Artificial Intelligence, machines are now capable of performing tasks that traditionally required human intelligence. AI systems can analyze large amounts of data quickly and produce decisions with high speed and consistency. Today, AI-based decisionmaking systems are used in healthcare diagnosis, autonomous vehicles, online recommendation systems, banking fraud detection, recruitment systems, and educational platforms. This project focuses on comparing AI and human decision-making behaviors using data analysis and machine learning techniques. The study aims to identify similarities and differences between human and AI responses in various decision-making situations.

Objectives of the Study:

- To compare AI and human decision-making patterns.
- To analyze response differences using statistical methods.
- To apply machine learning models for classification.
- To evaluate the ethical implications of AI-based decisions.
- To study legal and professional concerns related to AI systems.
- To understand how AI can support human decision-making instead of replacing it.

Scope of the Study:

This study focuses on analyzing collected response data from human participants and AIgenerated outputs. The project includes data preprocessing, exploratory data analysis, statistical testing, machine learning classification, and ethical evaluation.

Limitations:

- Limited number of participants.
- Responses may contain personal bias.
- AI responses depend on training data and prompts.
- Results may vary with larger or more diverse datasets.

3. Problem Statement / Case Background:

Artificial Intelligence systems are becoming more common in modern decision-making environments. Many organizations now use AI for recommendations, predictions, and automated judgments. However, there is still uncertainty regarding whether AI systems can truly match human reasoning abilities. Humans often make decisions based on emotional intelligence, ethics, empathy, creativity, and social understanding. In contrast, AI systems depend mainly on data patterns, algorithms, and learned statistical relationships. Although AI can process information quickly and reduce human workload, there are concerns regarding bias, fairness, explainability, and accountability. The main problem addressed in this case study is understanding how AI decision-making differs from human decision-making and identifying the strengths and weaknesses of both approaches.

Stakeholders Involved:

- Students and participants providing responses.
- AI systems generating decisions.
- Researchers and developers.
- Organizations implementing AI technologies.
- Regulatory and legal authorities.

Importance of the Case Study:

This case study is important because AI technologies are increasingly influencing critical sectors such as healthcare, banking, education, cybersecurity, and employment. Incorrect or biased AI decisions can create serious ethical, social, and legal consequences.

The study helps in understanding:

- Whether AI decisions are fair and reliable.
- The role of human judgment in sensitive situations.
- The ethical challenges associated with intelligent systems.
- The importance of explainable and transparent AI.

4. Literature Review / Theoretical Framework:

Artificial Intelligence refers to computer systems designed to perform tasks that normally require human intelligence. According to modern computing theories, AI systems can learn patterns from data and make predictions or decisions with minimal human intervention.

Researchers have studied AI decision-making in fields such as healthcare, finance, robotics, and cybersecurity. Previous studies show that AI systems are highly effective in data-driven environments but may struggle in emotionally sensitive or morally complex situations.

Human Decision-Making:

Human decision-making involves:

- Emotional intelligence
- Ethical reasoning
- Experience-based judgment
- Creativity and intuition
- Contextual understanding

Humans can adapt to unexpected situations more effectively than many AI systems.

AI Decision-Making:

AI systems mainly rely on:

- Historical data
- Machine learning algorithms

- Statistical analysis
- Pattern recognition
- Automated optimization

AI systems are faster and more scalable but can inherit biases from training data.

Related Professional Standards:

This study also considers ethical guidelines from:

- IEEE Code of Ethics
- ACM Code of Ethics
- BCS Code of Conduct

These standards emphasize fairness, privacy, transparency, accountability, and public welfare in computing systems.

5. Methodology / Approach:

The methodology of this project involved several structured phases including data collection, preprocessing, statistical analysis, machine learning implementation, and explainability analysis.

Data Collection:

Human Responses:

- Responses collected through Google Forms.
- Participants included students and individuals from different backgrounds.
- Questions focused on real-life decision-making situations.

AI Responses:

- AI-generated responses were collected using AI systems.
- The same decision-making scenarios were provided to the AI.

Data Preparation and Cleaning:

The following preprocessing steps were performed:

- Removal of duplicate entries.
- Handling missing values.
- Standardizing column names.
- Data merging and transformation.
- Label encoding of categorical values.
- Feature scaling and normalization.

Exploratory Data Analysis:

Several visualization and analysis techniques were applied:

- Summary statistics.
- Response distribution analysis.
- Scatter plot matrix.
- Boxplots and radar charts.
- Confusion matrix analysis.

Statistical Analysis:

The project used:

- T-Test
- Chi-Square Test
- Cohen's d Effect Size Analysis

Machine Learning Models Used:

- Logistic Regression
- Random Forest
- XGBoost

Tools and Technologies:

- Python
- Google Colab
- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Seaborn
- XGBoost

6. Analysis and Discussion:

Overview of Analytical Process

The analysis phase focused on identifying meaningful differences between AI-generated responses and human-generated responses using statistical analysis, exploratory data analysis, and machine learning techniques. The dataset was processed in Google Colab using Python libraries such as Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, and XGBoost.

The project workflow included:

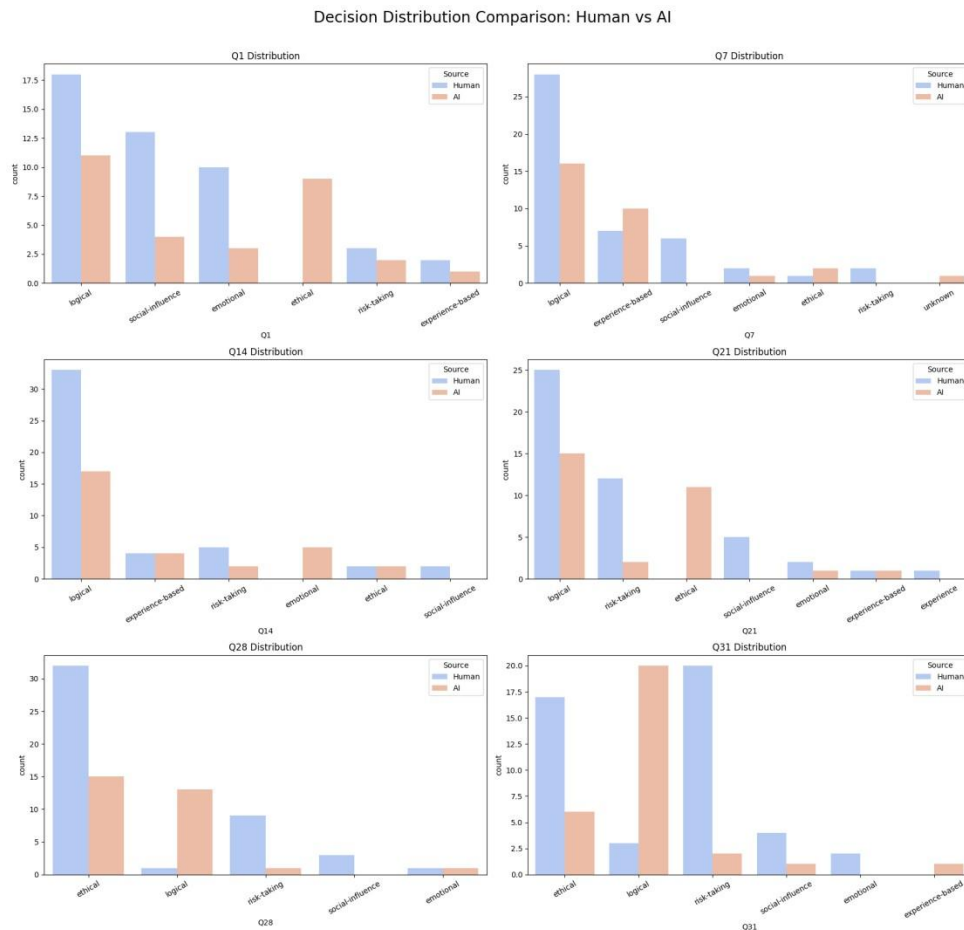
1. Data preprocessing and cleaning.
2. Exploratory data analysis (EDA).
3. Statistical testing.
4. Machine learning model training.
5. Model evaluation.
6. Explainable AI analysis.

Exploratory Data Analysis:

Several visualizations were generated during the analysis process to better understand the response distributions and patterns.

Boxplots:

Boxplots were used to compare AI and human response distributions. The results showed that human responses had higher variability because humans often make decisions based on emotions, experiences, and personal judgment. In contrast, AI-generated responses were more consistent because AI systems rely on learned patterns from training data.



Radar Charts:

Radar charts were used to compare multiple decision-making characteristics simultaneously. The charts showed that:

- Humans performed better in empathy and ethical reasoning.
- AI systems performed better in consistency and speed.
- Human responses showed stronger contextual understanding.



Statistical Analysis:

Statistical methods were used to determine whether differences between AI and human responses were statistically significant.

T-Test Analysis:

The independent T-Test was applied to compare the means of AI and human responses. The analysis revealed that several questions produced statistically significant differences between AI and human decision-making behavior. This indicates that humans and AI systems approach certain decision-making situations differently.

Chi-Square Test:

The Chi-Square test was used to evaluate relationships between categorical response variables.

The results showed noticeable associations between response type and decision-making category.

Cohen's d Effect Size:

Cohen's d analysis measured the strength of differences between groups.

The results indicated moderate to strong effect sizes for some decision-making scenarios, especially in questions involving ethics, emotions, and social reasoning.

Machine Learning Analysis:

Machine learning models were implemented to classify whether a response belonged to an AI system or a human participant.

Logistic Regression:

Logistic Regression served as the baseline model. The model provided moderate classification performance and helped establish a comparison benchmark.

Random Forest:

Random Forest improved prediction performance by capturing more complex relationships within the dataset.

Advantages observed:

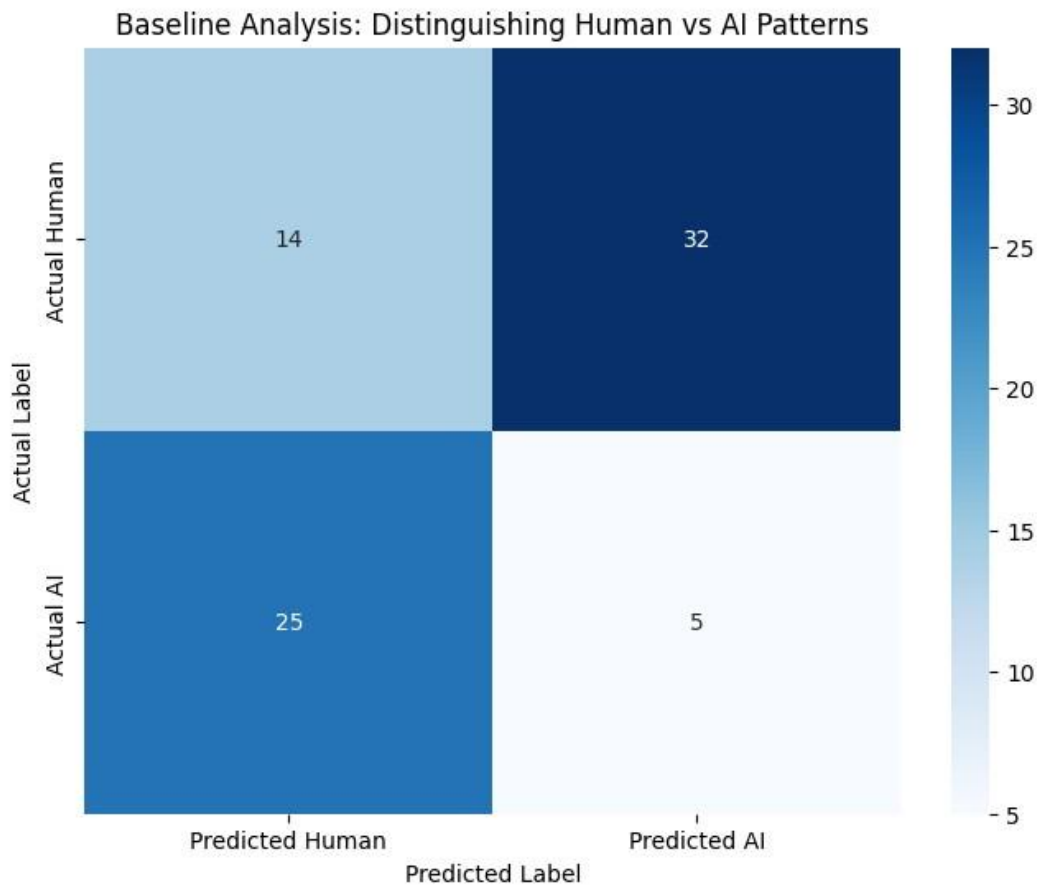
- Better handling of non-linear relationships.
- Improved classification accuracy.
- Reduced overfitting.

XGBoost:

XGBoost produced the best overall results among all implemented models.

The model achieved:

- Higher accuracy.
- Better feature learning.
- Stronger prediction consistency.



□

Explainable AI (XAI) Analysis:

One of the major goals of the project was improving AI transparency and interpretability.

Feature Importance Analysis:

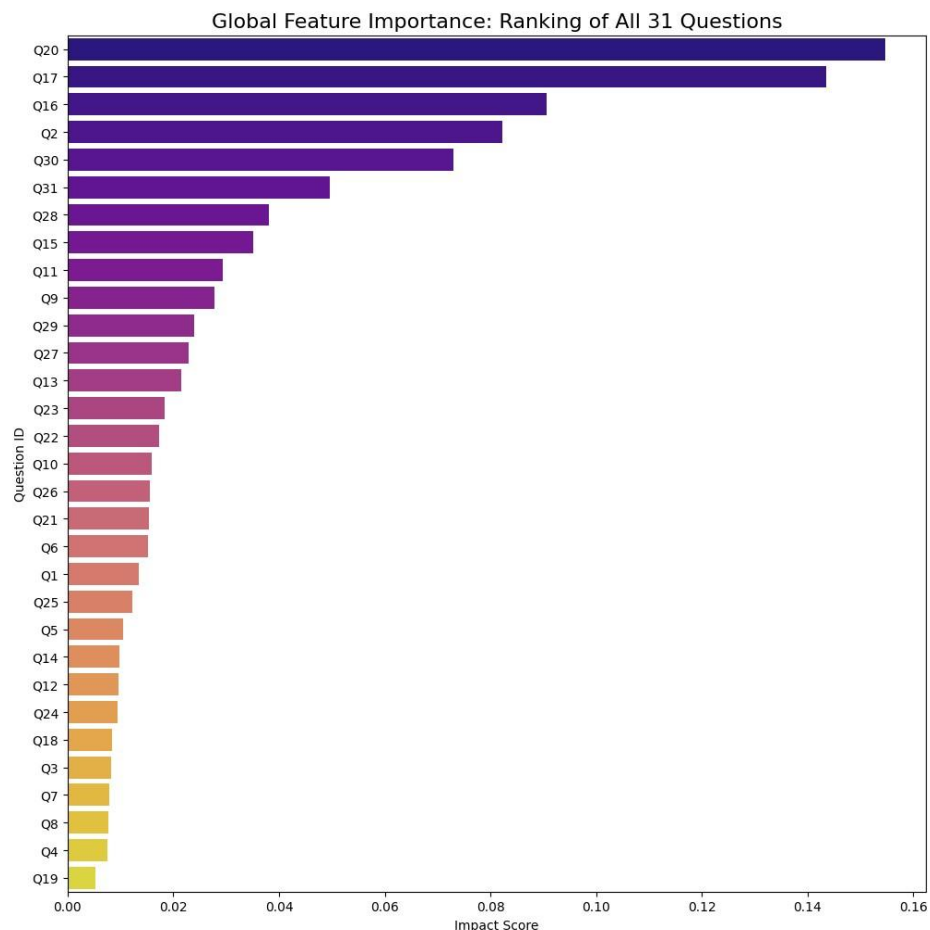
Feature importance analysis identified which questions most strongly influenced model predictions.

The analysis revealed that ethical and emotionally sensitive questions had greater importance when distinguishing AI responses from human responses.

SHAP Analysis:

SHAP (SHapley Additive Explanations) was used to interpret machine learning model behavior.

- How individual features affected predictions.
- Which variables increased or decreased classification probability.
- The internal reasoning patterns of the machine learning models.



Discussion of Findings:

The results of the project demonstrate that AI systems and humans have different strengths and weaknesses in decision-making.

Strengths of AI Decision-Making:

- Faster processing speed.
- Consistent responses.
- Efficient analysis of large datasets.

- Reduced manual workload.

Weaknesses of AI Decision-Making:

- Limited emotional understanding.
- Dependence on training data quality.
- Risk of algorithmic bias.
- Lack of deep contextual reasoning.

Strengths of Human Decision-Making:

- Ethical reasoning.
- Emotional intelligence.
- Creativity and adaptability.
- Contextual understanding.

Weaknesses of Human Decision-Making:

- Emotional bias.
- Slower processing.
- Inconsistency in repeated decisions.
- The overall findings suggest that AI systems should support human decision-making rather than fully replace it. Human oversight remains necessary in sensitive and ethical decision-making environments.

7. Ethical Analysis:

Ethics is one of the most critical aspects of AI-based systems. AI technologies can create serious societal impacts if they are not developed and used responsibly.

Ethical Issue 1: Algorithmic Bias

Relevant Code Clause:

- ACM Code 1.4: Be fair and take action not to discriminate.
- IEEE Principle 1: Hold paramount the safety, health, and welfare of the public.

Analysis:

AI systems learn from historical data. If the training data contains bias, the AI model may produce unfair or discriminatory decisions.

In this project, differences between AI and human responses highlighted how AI systems may prioritize patterns over ethical understanding.

Stakeholder Impact:

- Users may experience unfair treatment.
- Organizations may lose trust.
- Society may face discrimination risks.

Ethical Issue 2: Lack of Transparency**Relevant Code Clause:**

- ACM Code 2.5: Give comprehensive and thorough evaluations of computer systems.

Analysis:

Many AI systems operate as black boxes where users do not fully understand how decisions are made.

This project addressed transparency using explainable AI techniques such as SHAP analysis and feature importance visualization.

Stakeholder Impact:

- Reduced public trust.
- Difficulty in accountability.
- Challenges in debugging and auditing.

Ethical Issue 3: Privacy and Data Protection**Relevant Code Clause:**

- ACM Code 1.6: Respect privacy.
- BCS Code: Protect confidential information.

Analysis:

AI systems often require large amounts of personal data. Improper handling of data may violate user privacy.

The project emphasized responsible data handling and anonymized participant responses.

Stakeholder Impact:

- Risk of personal data exposure.
- Legal consequences for organizations.
- Loss of user confidence.

8. Legal and Regulatory Issues:

AI-based systems must comply with legal and regulatory standards to protect users and ensure responsible technology use.

PECA (Pakistan Electronic Crimes Act):

PECA addresses issues related to cybercrime, misuse of digital systems, and unauthorized data access. AI systems handling user data must ensure secure storage and responsible usage.

GDPR (General Data Protection Regulation)

GDPR emphasizes:

- Data protection.
- Right to privacy.
- Transparency in automated decision-making.

Intellectual Property Rights (IPR):

AI-generated content and algorithms may raise intellectual property concerns related to ownership, originality, and copyright.

Compliance Risks:

- Unauthorized data collection.
- Biased automated decisions.
- Lack of explainability.
- Weak cybersecurity measures.

Liability Issues:

If AI systems make harmful decisions, organizations may face legal responsibility for damages caused by inaccurate or biased outputs.

9. Recommendations / Proposed Solution:

Based on the findings of this study, the following recommendations are proposed:

Technical Recommendations:

- Improve training datasets to reduce bias.
- Use explainable AI techniques for transparency.
- Regularly audit AI systems for fairness.
- Implement stronger cybersecurity protections.

Professional Recommendations:

- Follow IEEE, ACM, and BCS ethical guidelines.
- Provide ethical AI training for developers.
- Ensure accountability in automated systems.
- Encourage responsible AI governance.

Proposed Hybrid Decision-Making Model

The study recommends a hybrid model where:

- AI handles data processing and predictions.
- Humans supervise ethical and sensitive decisions.
- Organizations maintain transparency and accountability.

10. Conclusion:

This case study explored the differences between AI and human decision-making using data analytics, statistical analysis, machine learning, and explainable AI techniques. The findings showed that AI systems are efficient, consistent, and effective for data-driven tasks. However, humans continue to perform better in situations involving ethics, emotions, empathy, and contextual reasoning. The project also highlighted important ethical and legal concerns related to fairness, bias, privacy, transparency, and accountability. Overall, the study concludes that AI should support human decision-making rather than completely replace it. Responsible AI implementation requires ethical governance, legal compliance, transparency, and continuous human oversight.

11. Personal Reflection:

This project provided valuable learning experience in data analytics, machine learning, ethical computing, and professional practices.

Through this project, I learned:

- How to clean and analyze real-world datasets.
- How machine learning models classify patterns.
- The importance of explainable AI.
- Ethical responsibilities in computing.
- The challenges of balancing technology with human values.

Challenges Faced

- Collecting and organizing reliable data.
- Handling missing and inconsistent responses.
- Understanding advanced machine learning concepts.

- Managing ethical analysis and professional reporting.

Skills Developed

- Python programming.
- Data visualization.
- Machine learning implementation.
- Critical thinking.
- Teamwork and communication.
- Professional documentation.

This project improved my understanding of workplace readiness and the professional responsibilities of computing professionals.

12. References:

1. Russell, S., & Norvig, P. Artificial Intelligence: A Modern Approach.
2. Ian Sommerville. Software Engineering.
3. ACM Code of Ethics and Professional Conduct.
4. IEEE Code of Ethics.
5. BCS Code of Conduct.
6. GDPR Official Documentation.
7. Pakistan Electronic Crimes Act (PECA).
8. Scikit-learn Documentation.
9. SHAP Explainability Documentation.
10. Research papers related to AI ethics and decision-making.

13. Appendices:

Appendix A — Sample Code Snippets

- Data preprocessing code.
- Machine learning model training code.
- Statistical testing implementation.

- Visualization scripts.

Appendix B — Visualizations

- Confusion matrices.
- Radar charts.
- Boxplots.
- Feature importance graphs.
- SHAP analysis figures.

Appendix C — Dataset Information

- Human response dataset.
- AI-generated response dataset.
- Question categories and labels.